Parameter-Efficient Transfer Learning for Remote Sensing Image Captioning

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***Abstract*— Remote sensing image captioning (RSIC) aims to generate accurate and concise textual descriptions for remote sensing (RS) images. It plays a significant role in the analysis of Earth observation data. The success of vision-and-language pretraining (VLP) models provides the foundation for their transfer to the RSIC task. To reduce the cost of transferring VLP models to downstream tasks, numerous parameter-efficient transfer learning (PETL) techniques have been proposed. However, most of them focus on fine-tuning general-purpose foundation models without fully considering the unique characteristics of RS data. In this article, we introduce PE-RSIC, a novel PETL framework tailored for RSIC. Specifically, the framework builds on a pretrained BLIP-2 model while further designing a lightweight cross-modal RS adapter (CRS-Adapter) and a Class Prompt. During training, all parameters of the pretrained model remain frozen, and the newly added CRS-Adapter modules are updated to efficiently transfer vision-and-language knowledge from the natural domain to the RS domain. The Class Prompt is obtained by projecting the vision-encoded [CLS] token into the decoder, guiding the model to generate more accurate captions. This approach enables the model to capture critical RS class features that might be lost during the query decoding process, with only a minimal increase in parameters. Extensive experiments show that our PE-RSIC framework outperforms full fine-tuning while using only 5% of the trainable parameters. The code will be available at https://github.com/Nick-Xzz/PE-RSIC.**

***Index Terms*— Image captioning, parameter-efficient transfer learning (PETL), remote sensing (RS).**

I. Introduction

THE continuous advancement of Earth observation technology makes obtaining multisource remote sensing (RS) data increasingly accessible. The tasks of extracting critical information from large-scale RS data remain a significant research focus in the RS field [1], [2], [3], [4]. Among these tasks, RS image captioning (RSIC) has emerged as a prominent research hotspot. The goal of RSIC is to accurately interpret and describe contextual objects in RS images using appropriate vocabulary, so that humans can intuitively grasp the key information contained within these images. This task holds significant application value across various fields, such as urban planning, disaster detection, military intelligence collection, and geographic information updating [5]. Traditional RSIC models typically use visual models for image feature extraction. During the text generation phase, language models are used to convert the visual feature vectors into the corresponding textual descriptions. Moreover, the recent success of large-scale vision-and-language pretraining (VLP) models [6], [7] has opened new possibilities for RSIC.

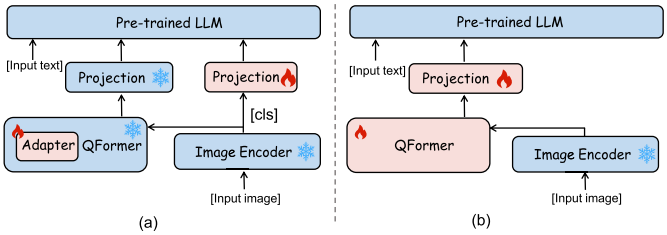


Fig. 1. Our approach builds upon BLIP-2 by augmenting QFormer with a set of trainable adapters. The learned query embeddings are merged with an additional encoded [CLS] token, providing the LLM with richer visual representations. (a) Ours. (b) BLIP-2.

VLP combines the robust representation ability of visual pretraining models with the excellent reasoning capabilities of large language models (LLMs). By leveraging pretraining on large-scale unlabeled data [8], these models exhibit substantial potential for multimodal representation and excellent transferability. Furthermore, parameter-efficient transfer learning (PETL) [9] has emerged as an effective strategy for adapting these models to specific vision–language downstream tasks. PETL typically achieves this by freezing the pretrained model and introducing lightweight trainable parameters, enabling efficient adaptation to downstream tasks while significantly reducing memory consumption. However, the existing methods primarily focus on downstream tasks in natural domains [10], which align more closely with pretraining data. This makes them suboptimal for the more complex RS domain, thereby limiting their application in RSIC. Therefore, there is a pressing need to develop PETL methods that can address the unique challenges of RSIC.

Specifically, PETL for RSIC faces the following challenges. First, VLP in natural scenes exhibit a significant domain gap with the RS domain. To bridge this gap, it is necessary to

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design a method with fewer trainable parameters that can efficiently explore the unique knowledge of RS image–text pairs while fully inheriting the prior knowledge structure learned by VLP in natural scenes. Second, compared with natural images, RS images exhibit high similarity between different object categories and are often influenced by complex geographical backgrounds. These factors make it difficult to extract target features effectively, which subsequently impacts the generation of accurate captions. Finally, RS images often display significant multiscale feature differences, with limited regions containing rich geospatial objects. Therefore, visual and textual semantic alignment is more challenging than in natural scenarios. Resolving this issue requires further designing stronger modal alignment mechanisms to improve caption generation ability.

To address the above issues, we propose a novel adapter-based framework for the RSIC task. Given the excellent representational capability of BLIP-2 in natural image captioning, we adopt it as the baseline. On the basis, we design a cross-modal RS adapter (CRS-Adapter) tailored to the RSIC task, enabling it to adapt to the unique characteristics of RS data and improve the alignment between visual and textual modalities. Based on extensive explorations, the design of our adapter includes the following key decisions.

1) Unlike traditional method of adding adapters after attention or FFN, we insert adapters in parallel within each transformer block of the QFormer and remove skip connections. This allows the adapter to more effectively learn domain differences while preserving the natural-domain prior knowledge of the pretrained transformer.
2) To better capture the complex features in RS, we insert two parallel branches after the multihead self-attention module. This maintains the integrity of query information in the QFormer while learning cross-attention mixed information.
3) To enhance modal alignment for RSIC, we incorporate a gating mechanism in the upsampling layer of the adapter. This mechanism adjusts the information flow, facilitating the interaction between different modal information and adaptively emphasizes the most relevant and useful cross-modal relationships for image caption.

We obtain a pretrained QFormer with an adapter that provides well-aligned RS visual embeddings for the subsequent frozen LLM text generator. However, these embeddings are constrained by the token count, which may lead to the loss of certain RS class-specific features. Classes form unique RS scenes and are directly relevant to image caption generation. To address this issue, we propose a Class Prompt generated from vision-encoded [CLS] tokens to better guide the LLM in generating RS captions, as shown in Fig. 1. Meanwhile, this approach requires minimal computation and memory overhead, which perfectly aligns with the lightweight nature of PETL methods. In conclusion, our contributions are as follows.

1) The CRS-Adapter is proposed, which efficiently learns RS domain knowledge while adaptively enhancing modal alignment and inheriting pretrained prior knowledge.
2) A Class Prompt is developed to help the model capture critical RS classes features, guiding the text decoder to produce more precise and effective RS captions.
3) We propose PE-RSIC, a novel and effective PETL framework for RSIC. Extensive experiments on mainstream datasets validate the effectiveness of our approach.

The remainder of this article is organized as follows. Section II reviews relevant studies. The proposed PE-RSIC approach is detailed in Section III. Section IV presents and analyzes the experimental results. Finally, the conclusion is given in Section V.

II. Related Work

A. RS Image Captioning

Since Qu et al. [11] introduce a pioneering multimodal architecture based on CNN–RNN frameworks for RSIC, numerous RSIC methods have emerged. Currently, mainstream approaches predominantly use a unified encoder–decoder framework [12]. This framework typically uses CNN-based architectures such as ResNet or transformer-based architectures such as vision transformer (ViT) as image encoders to extract rich RS visual features. The decoder usually uses structures such as RNN to translate visual embeddings into textual representations, thereby generating corresponding descriptive sentences. Moreover, some studies use transformers [13] as decoders to enhance the text generation capabilities of RSIC.

RS images contain a multitude of entities at various scales. To address this challenge, most methods apply attention mechanisms and feature aggregation strategies to enhance image feature representation [14]. Huang et al. [15] propose a multiscale feature aggregation network and designed denoising operations to eliminate redundant information from the aggregated features. Yuan et al. [16] propose a framework that combines multilevel attention with a multilabel attribute graph convolution to handle scale variations and visual relationships more effectively. Zhang et al. [17] introduce a label-attention mechanism to effectively leverage label information for guiding the computation of attention masks. In addition, some studies focus on improving language decoders to generate higher quality sentences. Li et al. [18] propose a multilevel attention model, adding three attention structures in the decoder to better mimic human attention, thereby enhancing the focus on both the images and sentences. Wang et al. [19] introduce a noun sequence generation decoder to enhance the capability of the model to understand key image targets. However, these methods do not fully leverage the advantages offered by the VLP paradigm.

Recently, some studies have applied VLP to RSIC. BITA [20] introduces an interactive Fourier transformer to replace the conventional transformer architecture, enabling more effective alignment of RS image–text features. RS-MoE [21] uses a Mixture of Experts framework, which uses dynamic task-specific prompts and LLMs to achieve higher scalability. However, these methods still require substantial computational resources, while PETL alleviates this limitation.

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B. Vision–Language Pretraining

In recent years, advances in NLP have driven the progress of VLP research. VLP has shown encouraging results in various downstream tasks [22], mainly including end-to-end vision–language and modular vision–language pretraining.

The end-to-end pretraining method is an early VLP paradigm. According to the encoder type, it can be categorized into dual-encoder architectures and fusion encoder architectures [23]. The fusion encoder processes image and text features as input and uses additional network branches for modality interaction. Among them, two-stream structures such as ViLBERT [24], ALBEF [25], and LXMERT [26] use two independent transformer-based branches for extracting region and text features, along with another transformer module for VL interaction. In contrast, OSCAR [27] and SimVLM [28] directly concatenate multimodal features and then input them into a single encoder. Dual-encoder (CLIP [29] and ALIGN [30]) after getting the image text embedding from the two encoders use some simpler and more direct VL alignment methods such as contrastive learning.

Due to the excellent feature extraction ability of large visual models (LVMs) [29], [31] and superior text generation ability of LLMs [32], the modular vision–language pretraining method freezes both the components and adds trainable modules to align visual features with the text space. The most representative ones are Flamingo [7] and BLIP-2 [6]. Specifically, Flamingo [7] injects visual features into the LLM generation process by cross-attention between the visual query and the layer newly inserted into the LLM, aligning arbitrarily interleaved visual and textual data sequences and achieving few-shot learning capabilities. BLIP-2 [6] uses a query transformer (QFormer) between frozen vision and language pretrained models to bridge the modality gap. InstructBLIP [33] introduces instruction tuning to BLIP-2, improving its zero-shot generalization capabilities. MiniGPT-v2 [34] further advances this paradigm through task-specific identifiers.

C. Parameter-Efficient Transfer Learning

PETL aims to adapt frozen pretrained large models to specific downstream tasks by updating a small number of parameters, thereby improving the memory consumption problem in pretrained models. It is first proposed in the field of NLP. Methods based on prompts [35] integrate vectors as prompts into the input to optimize the model. Based on CLIP, CoOp [36] first applies the learnable prompt to the CV field. However, since the generalization of the CoOp learned context is not insufficient, Zhou et al. [37] propose CoCoOp to extend CoOp by creating image-specific labels conditioned on the input. To combine the advantages of textual cue learning and visual cue learning, Wang et al. [38] propose a unified cue tuning (UPT) method by learning a micro neural network to jointly optimize prompts from different modalities.

Adapter-based methods [9] insert multiple adapters, each containing a small number of trainable parameters, into the model for fine-tuning. Recently, adapters have been widely used in CV and cross-modal fields. AdapterFormer [39] is an adapter of the original linear layer based on the ViT, which can be effectively adapted to vision tasks. Building on CLIP-Adapter [40], Zhang et al. [41] propose Tip-Adapter, an untrained approach to enhance CLIP's few-shot capability. Sung et al. [10] propose a cross-modal adapter for multimodal domains, which provides an early transfer learning framework for VL tasks. Recent studies [42] enhance cross-modal interaction by designing better adapter structures.

In addition to the above methods, low-rank adaptation (LoRA) [43] injects low-rank matrices into model weights, which reparameterizes the weight update matrix by the simple low-rank matrix. Zaken et al. [44] propose BitFit, a fine-tuning method that updates only the bias terms of the model. However, multimodal PETL of across domains remains underexplored. The existing PETL methods are suboptimal in the more challenging RS domain, as they are mainly tailored for downstream tasks with the same domain as pretraining.

III. Method

This section introduces the proposed PE-RSIC framework with its pipeline illustrated in Fig. 2. First, we describe the model architecture in Section III-A, followed by the CRS-Adapter in Section III-B. Finally, we introduce the Class Prompt, model training optimization, and caption generation in Sections III-C and III-D.

A. Model Architecture

Our model builds upon the fully pretrained BLIP-2 framework, comprising a pretrained vision encoder, an LLM, and a QFormer, further enhanced with the trainable CRS-Adapters and Class Prompt.

1) Vision Encoder: We use the ViT as the visual encoder to extract the visual representation of the input image and initialize its parameters with CLIP pretrained weights. Given an input image, we denote it as $I \in \mathbb{R}^{H\times W\times 3}$. First, I is divided into $N \times N$ patches V_{patch}, and then an additional classification token V_{cls} is added to the token sequence. In addition, ViT adds the positional embedding V_{pos} to each token to get the initial visual embedding sequence V_0, which can be expressed as

$$V_0 = \left[V_{\text{cls}}; V_{\text{patch}}\right] + V_{\text{pos}} \tag{1}$$

where V_{cls} denotes the added [CLS] token, V_{patch} denotes the patch tokens, and V_{pos} denotes the position embedding vector. The initial visual embedding sequence V_0 passes through L transformer blocks. The process of the Lth layer can be expressed as

$$\tilde{V}_l = \text{MSA}(\text{LN}(V_{l-1})) + V_{l-1} \tag{2}$$

$$V_l = \text{FFN}(\text{LN}(\tilde{V}_l)) + \tilde{V}_l \tag{3}$$

where $\text{MSA}(\cdot)$ represents the multihead self-attention module, $\text{LN}(\cdot)$ represents layer normalization, and $\text{FFN}(\cdot)$ represents the feedforward network. $\tilde{V}_l$ denotes intermediate hidden features extracted by the $\text{MSA}(\cdot)$ module in the lth ViT block, while V_l is the final output of the Lth block in the ViT. The [CLS] token from V_l is regarded as the global representation of the input image. Finally, the visual embedding V_e is obtained by applying normalization and projection to the V_l.

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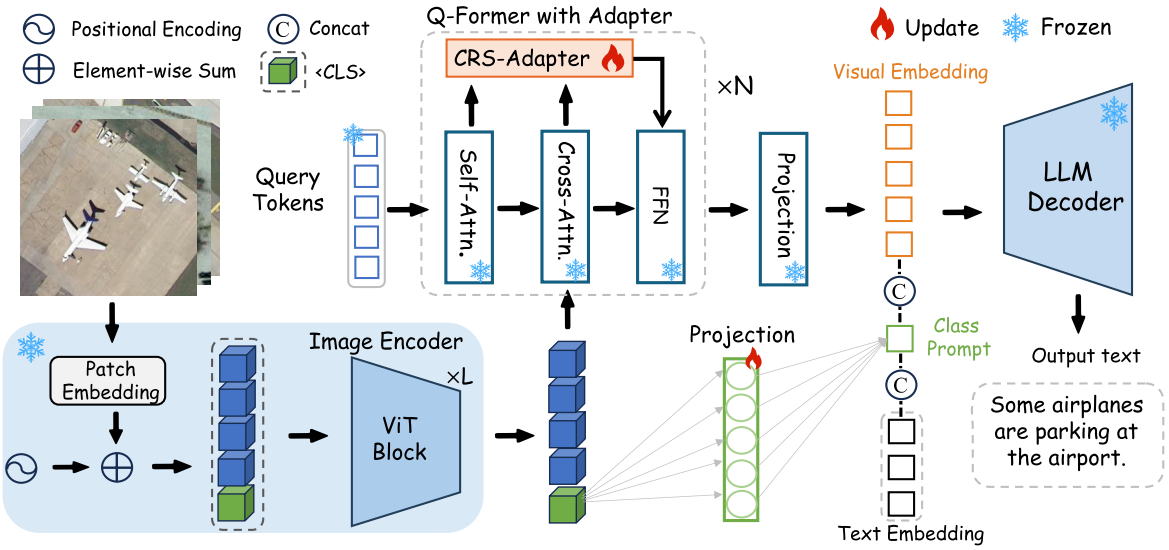


Fig. 2. Overall architecture of the proposed novel PE-RSIC framework, and it mainly consists of some key parts: the vision encoder, QFormer, CRS-Adapter, Class Prompt, and LLM decoder. The input image is processed by a pretrained ViT to extract visual features, which interact with the Q-Former integrated with a CRS-Adapter to generate visual embeddings. Subsequently, the vision-encoded [CLS] token is projected as a class prompt into the LLM. Finally, the visual embeddings are fed into the LLM to produce the final image–text description.

2) QFormer With Adapter: QFormer is a transformer-based structure to bridge the gap between image encoders and LLMs. We initialize it with the QFormer parameters from the pretrained BLIP-2 model. The input of QFormer is a query embedding, which interacts with each other through self-attention layers. After stacking N-1 layers, the self-attention output feature F_q can be calculated, i.e.,

$$F_q = \text{MSA}(\text{LN}(F_{n-1})) + F_{n-1}. \tag{4}$$

The F_q then interacts with the visual embedding V_e obtained in the image encoder through the cross-attention layer (inserted into all other transformer blocks). This process can be expressed as

$$\tilde{F}_n = \text{MCA}(Q = F_q, K = V_e, V = V_e) + F_q \tag{5}$$

where MCA(·) represents the multihead cross-attention module. Specifically, in each cross-attention layer, the self-attention output features F_q serve as Q, while the visual embedding V_e serves as both K and V. The hidden state $\tilde{F}_n$ finally passes through FFN(·), producing the final output F_n, i.e.,

$$F_n = \text{FFN}(\text{LN}(\tilde{F}_n)) + \tilde{F}_n. \tag{6}$$

This output F_n then passes through a projection layer to generate a more refined visual representation of the input image. This representation provides the most relevant information and ensures better alignment with the frozen LLM text generator, facilitating improved caption generation.

Moreover, considering the differences between RS image–text data and natural domain data, we propose the **CRS-Adapter**, which learns the specialized knowledge of RS image–text. The specific details are introduced in Section III-B.

3) Large Language Model: We connect QFormer with adapter to the frozen LLM to obtain the generative language ability of LLM. Specifically, the output query F_n is linearly projected to match the dimension of the LLM's input embedding. The projected vision embedding V_E is then added to the input text embedding. V_E acts as a soft prompt to condition LLM to generate image caption according to the visual representation extracted by QFormer. Meanwhile, to enrich the decoder with more comprehensive visual knowledge while considering computational cost, we introduce a Class Prompt, which will be detailed in the Section III-C.

B. CRS-Adapter

Building on the success of adapters [9] in NLP, an increasing number of adapter-based PETL approaches have demonstrated strong performance in downstream tasks. The adapter uses a downprojection linear layer to map the input features to a lower dimensional representation, followed by a nonlinear activation function (typically ReLU), and then an up-projection linear layer to restore the dimension. Given the input $X \in \mathbb{R}^{n\times d}$, this process can be expressed as

$$\hat{X} = f(X \cdot W_{\text{down}})W_{\text{up}} + X \tag{7}$$

where $W_{\text{down}} \in \mathbb{R}^{d\times\hat{d}}$ represents the downsampling matrix, $W_{\text{up}} \in \mathbb{R}^{\hat{d}\times d}$ represents the upsampling matrix, $f(\cdot)$ represents the nonlinear activation function, and $\hat{X} \in \mathbb{R}^{n\times d}$ represents the adapter features. To more effectively learn RS visual–text features without altering the pretrained model structure, we propose a dual-branch parallel CRS-Adapter. It uses weight-sharing and gating mechanism to control information flow, enhancing the ability of model to learn relevant multimodal relationships, as shown in Fig. 3.

1) Residual Learning for Adapter: In most scenarios, the adapter is inserted after FFN, and (6) can be modified as follows:

$$F_n = \text{Adapter}(\text{FFN}(\text{LN}(\tilde{F}_n)) + \tilde{F}_n \tag{8}$$

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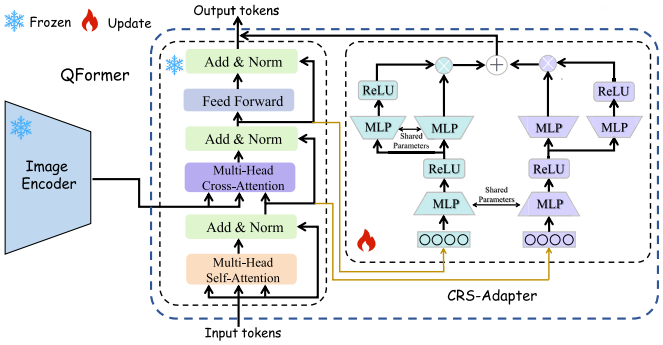


Fig. 3. Implementation details of our CRS-Adapter. It takes inputs from the MCA(·) and MSA(·) modules and processes them through parallel branches. After processing, the features are combined and then output.

where F_n represents the output of the Nth layer of QFormer. The above methods have been proven to be effective in natural scenes. However, to adapt the pretrained VLP to RS data while inheriting the prior knowledge structure of natural scenes, we insert adapters in parallel and remove skip connections. This allows the adapter to learn domain differences more efficiently without changing the pretrained transformers. First, we insert adapter in parallel after MCA(·). This added branch learns RS information extracted by the MCA(·). It helps bridge the domain gap between the natural scene pretrained model and RS images, enabling the model to acquire more representative image features. Furthermore, the cross-attention process in QFormer may destroy the integrity of the query information. Therefore, we further introduce residual learning of the query to maintain the information by introducing additional adapters. Given (4) and (5), (8) can be modified as

$$F_l = \text{FFN}(\text{LN}(\tilde{F}_l) + \text{Adapter}(\tilde{F}_l) + \text{Adapter}(F_q)). \tag{9}$$

2) Gating Mechanism With Knowledge Sharing: Introducing a query adapter may lead to additional parameter updates, which we do not expect from a lightweight perspective. To minimize additional parameter costs, we implement a weight-sharing mechanism specifically in the downsampling layer of the two adapter branches. Surprisingly, this weight-sharing mechanism even leads to better performance. This is attributed to weight-sharing facilitating the fusion of diverse information. As shown in Fig. 3, the input feature vector is projected to a smaller dimension and then passed through an ReLU activation function. Inspired by the success of gated linear units in the feedforward layers of transformers [45], we introduce a gating mechanism in the up-projection layer. Specifically, we compute the elementwise product of two linear transformations, denoted as W_u and W_g, where W_u is activated by ReLU. This gating mechanism helps the adapter control the information flow, potentially emphasizing the most useful and relevant multimodal relationships. Specifically, we denote the output of MSA(·) at the Nth layer $F_q \in \mathbb{R}^{n\times d}$ as $X^{\mathcal{T}}$ and denote the output of the MCA(·) at the Lth layer $\tilde{F}_n \in \mathbb{R}^{n\times d}$ as X^S. The adaptation process can be written as

$$X^{\mathcal{T}}_{\text{down}} = \text{ReLU}\left(X^{\mathcal{T}} \cdot W_d^{\text{share}}\right) \tag{10}$$

$$X^{\mathcal{T}}_{\text{Adapter}} = \text{ReLU}(X^{\mathcal{T}}_{\text{down}} \cdot W_u) \otimes X^{\mathcal{T}}_{\text{down}} \cdot W_g \tag{11}$$

$$X^{S}_{\text{down}} = \text{ReLU}\left(X^{S} \cdot W_d^{\text{share}}\right) \tag{12}$$

$$X^{S}_{\text{Adapter}} = \text{ReLU}(X^{S}_{\text{down}} \cdot W_u^{\text{share}}) \otimes X^{S}_{\text{down}} \cdot W_u^{\text{share}} \tag{13}$$

$$X_{\text{Adapter}} = X^{S}_{\text{Adapter}} + X^{\mathcal{T}}_{\text{Adapter}} \tag{14}$$

where X_{Adapter} represents the output of the CRS-Adapter, which is added to the FFN output.

C. Class Prompt

Due to the token number constraint in QFormer, some features of RS entity targets (i.e., classes) may be lost during feature alignment. These classes define unique scene semantics, which are critical for image caption generation. Therefore, we propose a class-aware prompt generated by projecting the vision-encoded [CLS] token. This prompt helps guide the model in capturing geospatial targets and understanding visual representations, improving the ability of LLM to generate RS captions. Specifically, we extract the [CLS] token V_{cls} from the output V of the vision encoder, and this process can be expressed as

$$V_{\text{cls}} = V[\text{cls}]. \tag{15}$$

Then, the [CLS] token V_{cls} is fed into a fully connected projection layer to align its dimensions with the input embeddings of the LLM decoder

$$C_{\text{prompt}} = \text{MLP}(\text{V}_{\text{cls}}). \tag{16}$$

Finally, the projected Class Prompt C_{prompt} is integrated into the LLM.

D. Training Optimization and Caption Generation

During training, images are input to the ViT to extract visual features, which interact with the CRS-Adapter-integrated QFormer to produce visual embeddings V_e. Subsequently, a fully connected layer linearly projects the vision-encoded [CLS] token to match the input dimensionality of the LLM, forming the Class Prompt C_{prompt}. The projected visual embeddings V_E and C_{prompt} are then concatenated as a prefix to the input text embeddings to generate the final image caption. Only the Class Prompt and CRS-Adapter are trainable during training, while all other components remain frozen. To optimize the model, we use cross-entropy loss $\mathcal{L}_{\text{lm}}$, widely used for LLMs. This loss measures the difference between the predicted token probabilities and the ground-truth sequence. Specifically, the loss function is defined as

$$\mathcal{L}_{\text{lm}} = -\sum_{t=1}^{T} \log(p_\theta(y_t^* \mid y_{1:t-1}^*)) \tag{17}$$

where $y_{1:t-1}^*$ denotes the ground-truth tokens preceding step t, T is the sequence length, and θ represents the model parameters. This loss function drives the model to generate accurate captions by reducing the gap between the predicted and actual token distributions. During inference, we directly load the pretrained model along with our trained adapter and projection layer parameters. In addition, beam search is used to improve the diversity of the generated captions to a certain extent.

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IV. Experiments

This section begins by presenting the datasets and evaluation metrics used by the algorithm. Subsequently, the specific details of the implementation are provided. Meanwhile, we assess the algorithm's performance on three datasets and present several visualization examples. Finally, the findings from the ablation experiments are discussed.

A. Datasets

To evaluate the performance of the proposed method, experiments are performed on the UCM, NWPU, and RSICD datasets. A detailed explanation is provided below.

1) *UCM-Caption Dataset [11]:* This dataset includes 2100 images categorized into 21 land-use classes, including agricultural, aerial, beach, and dense residential. Each image has a resolution of 256 × 256 pixels and is provided in RGB format. In total, the dataset contains 10 500 unique annotations. The UCM-Caption Dataset is widely used for aerial image captioning research.
2) *NWPU-Caption Dataset [48]:* This dataset contains 31 500 aerial images across 45 land-use categories, including airport, bridge, desert, and farmland. Each image is 256 × 256 pixels in size. The dataset includes five unique captions for each image, resulting in 157 500 human-written descriptions.
3) *RSICD Dataset [55]:* This dataset comprises 10 921 RS images sourced from different platforms. These images cover 30 land-use categories, such as airport, desert, farmland, and playground. Each image has a resolution of 224 × 224 pixels with some variations. To maintain annotation consistency, the dataset provides up to five captions per image, resulting in 54 605 sentences.

B. Evaluation Metrics

It is essential to evaluate the quality of generated captions for RSIC, and a variety of metrics are commonly used in image captioning. In this study, we use BLEU1-4, METEOR, ROUGE_L, CIDEr, and SPICE to comprehensively assess caption performance.

1) BLEU measures the n-gram overlap between generated and reference texts, focusing on precision. It evaluates n-grams for $n = 1, 2, 3, 4$ (BLEU-1 to BLEU-4), making it effective for tasks such as machine translation and captioning.
2) METEOR combines exact word matching with stemming and synonym recognition. It calculates a weighted harmonic mean of precision and recall, capturing semantic relationships better than n-gram-based metrics.
3) ROUGE_L evaluates similarity using the longest common subsequence (LCS). It emphasizes recall and is suitable for assessing content overlap in summarization and captioning.
4) CIDEr uses TF-IDF weighting to highlight n-grams that are both significant and frequently shared across multiple reference captions, ensuring that the generated captions closely align with the consensus of human annotations.
5) SPICE evaluates semantic content by comparing objects, attributes, and relationships. It uses scene graphs to assess deeper meaning beyond surface-level text similarity.

C. Experimental Settings

1) Data Setting: For fair comparison to the existing methods, each dataset is split into three subsets: 80% of the image–caption pairs are used for training, 10% for validation, and the remaining 10% for testing. During preprocessing, image sizes are resized to 224 × 224.

2) Frozen Pretrained Model: We use the ViT-g/14 from EVA-CLIP [56] as the visual encoder. For the language model, we adopt OPT-2.7B[1] as the decoder-only LLM, directly loading its pretrained weights. The QFormer[2] is initialized with fully pretrained BERT parameters from BLIP-2.

3) Fine-Tuning Setting: Fine-tuning is performed using the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$, and a weight decay of 0.05. The training process spans 40 epochs, starting with a linear learning rate warm-up phase where the learning rate gradually increases from 1×10^{-6} to 1×10^{-4}. After this warm-up phase, a cosine learning rate decay strategy is applied, with the minimum learning rate set to 0. Mixed precision training is used to enhance computational efficiency, and the training and validation batch sizes are configured as 64 and 32, respectively. During inference, we apply a beam search algorithm to achieve a balance between computational efficiency and prediction accuracy. All the experiments are implemented on an NVIDIA GeForce RTX 4090 GPU.

D. Comparison to Existing Methods

To evaluate the performance of the proposed method, we conduct comparisons with several mainstream approaches, all of which are based on encoder–decoder architectures. These include both the traditional methods and pretrained transformer-based approaches.

1) Traditional Methods: We first compare our proposed method with several advanced traditional RSIC methods, including Word Sentence [46], GVFGA+LSGA [47], GLCM [49], MLCA-Net [48], MLAT [50], and HCNet [51]. These methods use various attention mechanisms and feature integration strategies. A brief overview is as follows. Word Sentence [46] extracts key terms from images and constructs sentences through sequencing tasks. GVFGA+LSGA [47] uses two attention mechanisms to reduce redundant visuals and enhance VL feature integration. GLCM [49] combines global and local features, enhancing generation and interpretability by linking words to relevant visual features. MLCA-Net [48] resolves scale variation and category ambiguity through multilevel and contextual attention. MLAT [50] integrates multiscale features from convolutional layers and uses LSTM to aggregate information, refining features before passing to a transformer decoder for sentence generation. HCNet [51] uses a hierarchical feature aggregation module and designs a cross-module feature interaction module to reduce the modality gap between text and image features.

[1]Pretrained weights for OPT 2.7B.
[2]Pretrained weights for BLIP-2 QFormer.

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2) Pretrained Transformer-Based Methods: To fully evaluate our performance, we select several RSIC methods based on pretrained transformers, as described below. ClipCap [52] uses a mapping network to convert CLIP embeddings into prefixes, which are fed into a fine-tuned GPT-2, allowing efficient image captioning without extra pretraining. PureT [53] uses SwinTransformer for grid-level feature extraction and integrates a refining encoder and decoder in a transformer-based framework. RSGPT [54] fine-tunes InstructBLIP on high-quality human-annotated RSIC datasets for enhancing performance. In addition, we compare with the baseline BLIP-2 and its extension BITA to validate our effective use of the VLP paradigm. BLIP-2 [6] serves as our baseline, introducing a QFormer to bridge the modal gap. After two stages of pretraining, the QFormer generates visual representations interpretable by the LLM. BITA [20] builds upon BLIP-2 by introducing interactive Fourier transformer to replace the traditional transformer, enabling more effective alignment of RS image–text features.

E. Experimental Results and Analysis

1) Quantitative Comparison: Tables I–III show the quantitative results of different methods on the UCM-caption, NWPU-caption, and RSICD datasets. It is evident that our method outperforms other comparison methods, benefiting from the VLP paradigm, which combines the strong representation and reasoning capabilities of visual pretraining models and LLMs. Furthermore, our proposed CRS-Adapter effectively learns domain-specific knowledge in RS and bridges the multimodal gap between the visual pretraining model and the LLM more efficiently.

The results from the UCM-caption dataset show that traditional methods (e.g., GVFGA+LSGA, MLCA-Net, and GLCM) consistently underperform across multiple metrics compared with methods using transformers as both the encoder and decoder (e.g., PureT and BLIP-2). In addition, the HCNet demonstrates significant performance improvements over traditional methods due to the integration of an additional feature alignment module. The non-RS method BLIP-2 leverages the powerful representation and reasoning capabilities of a pretrained visual encoder and LLM. By introducing interactions between visual and textual features through QFormer, it achieves significantly better results than most advanced comparison methods across various metrics. This highlights the superiority of the VLP paradigm. Our approach further enhances QFormer by incorporating a trainable adapter that can bridge the domain gap while retaining prior knowledge from the natural domain, leading to more robust RS visual representations. A related approach, BITA, similarly builds on BLIP-2 by proposing the IFT module. BITA outperforms PureT and BLIP-2 on a range of metrics. Compared with BITA, our method achieves better results, with our CIDEr score improving by approximately ten points over theirs, and demonstrating significant advantages in parameter efficiency. The results on both NWPU and RSICD datasets are similar to the UCM dataset, with transformer-based methods outperforming traditional approaches. Among the transformer-based approaches, those adopting the VLP paradigm demonstrate superior performance. Our method also achieves highly competitive results.

2) Qualitative Comparison: Fig. 4 shows some typical RS scenes from three experimental datasets, along with the corresponding captions generated by three different methods: HCNet, BLIP-2 (base), and PE-RSIC. Clearly, compared with HCNet and BLIP-2, the caption generation results of our method are superior in terms of word accuracy, sentence comprehensiveness, and semantic consistency. Specifically, in Fig. 4(b) and (c), our model not only provides a complete description of the images but also generates detailed descriptions of the spatial arrangement of objects, enriching sentence detail and enhancing the sense of the scene. Likewise, in Fig. 4(f), our method accurately describes not only the colors of the beach and the waves but also their positional relationship, whereas some methods even produce incorrect descriptions. In Fig. 4(e), while other methods fail to capture the term "terminal" accurately, our approach not only correctly identifies the number and size of the aircraft but also demonstrates strong scene perception capabilities, offering a more comprehensive depiction of the airport environment. Moreover, compared with other methods, our model delivers more detailed object descriptions, such as "tall buildings" and "green trees" in Fig. 4(g) and "farmlands of the same size" in Fig. 4(j). Notably, our method continues to perform well in complex scenes. For example, in Fig. 4(d), our approach identifies"bunkers," which are missed by other methods and even the ground-truth labels. Similarly, in Fig. 4(h), PE-RSIC successfully highlights the feature "empty parking spaces." Finally, in Fig. 4(l), our method is the only one to describe all the objects in the scene.

3) Trained Parameter Efficiency: Fig. 5(a) illustrates the relationship between parameter count and CIDEr score during training across different methods. Evidently, our approach achieves the best performance while requiring the minimal trainable parameters compared with other methods. Specifically, our model only requires 9.6M parameters, representing a 95% reduction in trainable parameters compared with full fine-tuning. Notably, CRS-Adapter accounts for 5.9M parameters (with a bottleneck dimension $r = 128$), and Class Prompt accounts for 3.6M parameters. The proposed PETL framework achieves remarkable efficiency and effectiveness in RSIC tasks, showcasing the viability of adapting VL pretrained models from the natural domain to address challenges in the RS domain.

F. Limitation

Despite the promising results achieved by PE-RSIC across multiple benchmark datasets, the proposed framework has certain limitations in handling RS images containing small objects. Specifically, the model sometimes fails to accurately describe fine-grained objects that occupy only a small portion of the image. As shown in Fig. 6, a vehicle is small and embedded within complex urban backgrounds, which make it difficult for our model to accurately describe them. Our current method uses a global [CLS] token extracted from the vision encoder to generate a Class Prompt. It lacks the spatial granularity to focus on fine or localized targets. To address

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TABLE I

EXPERIMENTAL RESULTS OF THE UCM DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS, RESPECTIVELY. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED. THE GRAY SHADING REPRESENTS THE BASELINE

Method	Params(M)	BLEU-l	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Traditional methods									
Word-Sentence [46]	13	79.31	72.37	66.71	62.02	43.95	71.32	278.71	-
GVFGA+LSGA [47]	-	83.19	76.67	71.03	65.96	44.36	78.45	332.70	48.53
MLCANet [48]	-	82.6	77.0	71.7	66.8	43.5	77.2	324.0	47.3
GLCM [49]	10	81.82	75.40	69.86	64.68	46.19	75.24	302.79	-
MLAT∗ [50]	161	82.26	75.39	69.59	64.24	42.68	77.29	308.26	-
HCNet [51]	258	88.26	83.35	**78.85**	**74.49**	48.65	83.91	351.83	-
Transformer-based methods									
Clipcap∗ [52]	32	82.13	74.88	68.72	63.20	42.32	77.42	323.91	-
PureT [53]	220	85.73	80.20	75.62	71.29	46.86	82.01	349.00	47.94
BLIP-2 [6]	188	88.04	82.23	76.89	71.84	47.32	83.36	380.31	54.78
BITA [20]	171	88.89	83.12	77.30	71.87	46.88	83.76	384.50	54.88
RSGPT [54]	-	86.12	79.14	72.31	65.74	42.21	78.34	333.23	-
Ours	9.6	**90.03**	**84.08**	78.17	74.33	**49.33**	**84.72**	**394.57**	**55.79**

TABLE II

EXPERIMENTAL RESULTS OF THE NWPU DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS, RESPECTIVELY. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED. THE GRAY SHADING REPRESENTS THE BASELINE

Method	Params(M)	BLEU-l	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Traditional methods									
MLCANet [48]	-	74.5	62.4	54.1	47.8	33.7	60.1	126.4	28.5
MLAT∗ [50]	161	85.27	76.74	70.07	64.79	43.27	74.96	185.56	-
Transformer-based methods									
Clipcap∗ [52]	32	83.94	74.21	66.19	59.54	41.42	73.85	172.65	-
PureT [53]	220	**88.82**	80.31	73.30	67.50	42.32	75.84	195.12	27.43
BLIP-2 [6]	188	87.78	80.24	73.51	67.62	45.01	78.44	193.86	33.28
BITA [20]	171	88.54	**80.70**	73.76	67.60	45.27	**78.53**	197.04	33.65
Ours	9.6	88.47	79.69	**74.15**	**67.66**	**45.35**	78.14	**199.74**	**33.98**

TABLE III

EXPERIMENTAL RESULTS OF THE RSICD DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS, RESPECTIVELY. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED. THE GRAY SHADING REPRESENTS THE BASELINE

Method	Params(M)	BLEU-l	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Traditional methods									
Word-Sentence [46]	13	72.40	58.61	49.33	42.50	31.97	62.60	206.29	-
GVFGA+LSGA [47]	-	67.79	56.00	47.81	41.65	32.85	59.29	260.12	46.83
MLCANet [48]	-	75.70	63.40	53.90	46.10	35.10	64.60	235.60	44.40
GLCM [49]	10	77.67	64.92	56.42	49.37	36.27	67.79	254.91	-
Transformer-based methods									
PureT [53]	220	77.05	65.75	56.63	49.19	37.66	67.41	275.56	48.87
BLIP-2 [6]	188	76.91	66.04	57.02	49.60	40.28	69.86	294.66	53.11
BITA [20]	171	77.38	66.54	57.65	50.36	**41.99**	71.74	304.53	**54.79**
Ours	9.6	**78.20**	**67.85**	**58.74**	**51.07**	41.02	**72.14**	**304.62**	54.16

this, future work may consider incorporating finer grained object cues while avoiding excessive computational overhead to enhance the sensitivity of model to small object.

G. Ablation Study

In this section, we design ablation in the UCM-Caption and NWPU-Caption datasets to demonstrate the effectiveness of the design module.

1) Comparison to Traditional PEFT Methods: As shown in Table IV, our method significantly outperforms traditional PEFT techniques with only a slight increase in parameter count. Specifically, in terms of CIDEr, +18.36 over Bitfit, +19.34 over Adapter, and +12.26 over LoRA, with notable advantages in other metrics as well. This demonstrates the superiority of our framework over conventional PEFT methods. Unlike PEFT techniques designed for natural domains, our framework is specifically tailored to address the unique challenges of complex visual feature analysis and modality alignment in the RS domain, specifically designed for RSIC.

2) Bottleneck Dimension and BeamSize: We experiment with different bottleneck dimensions of CRS-Adapter and beam sizes during inference. From Fig. 5(b), we observe that as the bottleneck dimension increases, both the parameter size and model performance improve. However, when the embedding dimension reaches 128, the model performance begins to decline. This may be because larger embedding

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UCM

(a)



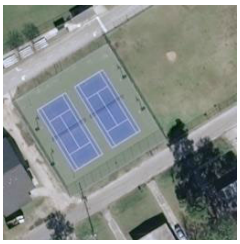
GT: Two straight freeways in the desert with no car on them.
HCNet: Two straight freeways in the desert.
Base: There are two straight freeways in the desert with no cars.
Ours: There are two straight freeways in the desert with no cars on them.

(b)



GT: Lots of boats docked at the harbor and the water is deep blue.
HCNet: Many boats docked at the harbor.
Base: Lots of boats docked at the harbor and the water is blue.
Ours: Lots of boats docked **neatly** at the harbor and the water is deep blue.

(c)



GT: There are two tennis courts on the lawn and surrounded by some plants
HCNet: Two tennis courts arranged neatly.
Base: There are two tennis courts arranged neatly on the lawn
Ours: There are two tennis courts arranged **neatly** and surrounded by some plants.

(d)



GT: A part of a golf course with green turfs and some trees.
HCNet: A golf course with grass and some trees.
Base: This is part of a golf course with turfs and trees..
Ours: This is a part of a golf course with green turfs and **some bunkers** and trees.

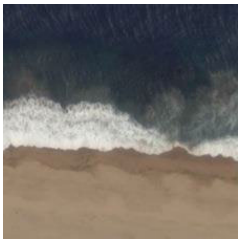
RSICD

(e)



GT: Five planes are near a large building in an airport
HCNet: Some planes are close to a big building in the airport.
Base: There are five planes near a large building at the airport.
Ours: Five planes are parked near **a large terminal** in an airport.

(f)



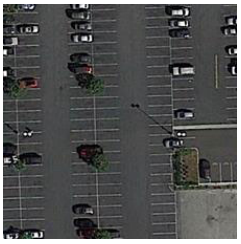
GT: White waves in green ocean is near yellow beach
HCNet: A green lawn is near a piece of yellow beach.
Base: White waves in a green ocean near a yellow beach
Ours: White waves are **between** green ocean and yellow beach.

(g)



GT: Many buildings are in a commercial area
HCNet: There are many buildings in a commercial area.
Base: Many buildings are some trees located in a business area
Ours: Many **tall** buildings and **some green trees** are in a commercial area.

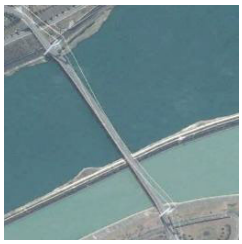
(h)



GT: some cars are parked in a parking lot with several green trees.
HCNet: Some cars are parked in a parking area with a few trees.
Base: Several cars are parked in a parking lot with green trees
Ours: Some cars are parked in a parking lot with **several empty parking spaces** and green trees

NWPU

(i)



GT: A slender, Straight bridge connecting the land on both sides of the blue waters.
HCNet: A bridge connects the land on both sides of the blue water.
Base: A slender, straight bridge connects the land across the blue water
Ours: A slender, Straight bridge connects the land on both sides of the turquoise water

(j)



GT: Several dark green circular farmlands are neatly arranged on the ground.
HCNet: Several dark green circular farmlands are neatly placed on the ground.
Base: There are some dark green circular farmlands arranged neatly.
Ours: Some dark green circular farmlands of **the same size** are neatly arranged on the ground.

(k)



GT: The cumulus clouds are located above the land covered by dense vegetation
HCNet: The clouds are above the land with a lot of vegetation.
Base: The stratus clouds are located above the land covered by a lot of vegetation.
Ours: The stratus clouds are located above the land covered by dense vegetation.

(l)

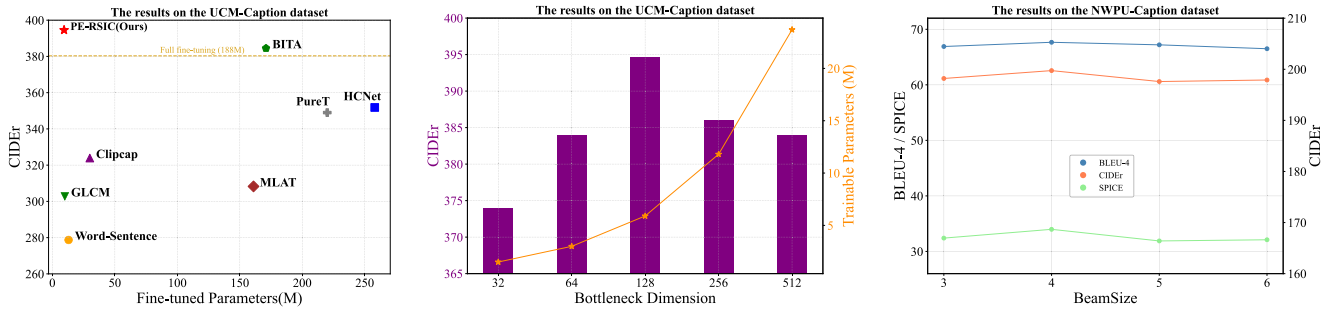


GT: The mobile home park has some dense mobile homes and a swimming pool and some trees are among these mobile homes
HCNet: There are many buildings and trees in the mobile home park
Base: The mobile home park has some dense mobile homes and some trees.
Ours: The mobile home park has some dense mobile homes arranged in rows and some trees and a swimming pool.

Fig. 4. Examples of sentences generated by PE-RSIC from the UCM dataset [(a)–(d)], the RSICD dataset [(e)–(h)], and the NWPU dataset [(i)–(l)]. The red words and blue words indicate the disadvantages of other methods and the advantages of our method, respectively.

dimensions lead to overfitting. Therefore, we choose the bottleneck dimension of our adapter to be 128. As shown in Fig. 5(c), the model achieves its highest BLEU-4, CIDEr, and SPICE scores when the beam size is equal to 4, with

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(a) (b) (c)

Fig. 5. (a) Comparison of PE-RSIC with different models in terms of CIDEr performance versus the number of trainable parameters. (b) Bottleneck dimension: the impact of different bottleneck dimensions on the model performance CIDEr and trainable parameters (M). (c) Beam size: the impact of different beam sizes on the model performance BELU-4, SPICE, and CIDEr.

TABLE IV

ABLATION RESULTS ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Method	Params(M)	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Full Fine-Tuning	188.0	88.04	82.23	76.89	71.84	47.32	83.36	380.31	54.78
Adapter	9.6	82.45	78.32	72.89	68.57	45.12	79.86	375.23	53.14
LoRA	7.1	87.78	81.93	76.12	72.56	47.89	82.84	382.31	54.35
BitFit	3.8	87.21	80.85	76.48	71.24	45.04	81.93	376.21	53.24
Ours	9.6	**90.03**	**84.08**	**78.17**	**74.33**	**49.33**	**84.72**	**394.57**	**55.79**

TABLE V

ABLATION RESULTS ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Method	Baseline	CRS-Adapter	Class Prompt	Full Feature	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Model1	✓				88.04	82.23	76.89	71.84	47.32	83.36	380.31	54.78
Model2	✓	✓			89.07	83.62	78.50	73.57	47.91	84.53	389.26	54.84
Model3	✓	✓		✓	89.25	**84.12**	77.89	74.05	48.97	**84.84**	391.68	55.35
Model4(Ours)	✓	✓	✓		**90.03**	84.08	**78.17**	**74.33**	**49.33**	84.72	**394.57**	**55.79**

TABLE VI

ABLATION RESULTS ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Method	Resdual	Gating	Weight-Sharing	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
Model5				83.71	77.42	71.38	66.84	42.13	79.29	360.71	50.64
Model6	✓			88.64	82.02	77.53	72.16	46.23	82.31	377.62	53.41
Model7	✓		✓	89.32	83.61	78.14	73.07	47.11	83.54	388.46	54.06
Model4	✓	✓	✓	**90.03**	**84.08**	**78.17**	**74.33**	**49.33**	**84.72**	**394.57**	**55.79**



GT: A tan cross-shaped roof, The main house has a light blue round roof, A church with a tower, And buildings, Roads and cars around.

Ours: A tan cross-shaped roof, the main house with a light blue round roof, a church with a tall tower, and buildings and roads nearby.

Fig. 6. Example of sentence generated by PE-RSIC compared with GT.

performance decreasing as the beam size deviates from this value.

3) Effectiveness of Main Components: As shown in Table V, to verify the effectiveness of each component, we conduct ablation experiments by progressively incorporating components into the baseline BLIP-2 until the complete PE-RSIC framework is established. Compared with full fine-tuning, introducing the CRS-Adapter component alone already leads to a certain degree of performance improvement. When both the components are used together, the CIDEr score increases by 14.26, and the BLEU-4 score increases by 2.49. We compared the performance when projecting the [CLS] token versus the full feature into the LLM. Model 3 represents the projection of the full feature from the penultimate layer to the LLM with a fully connected layer to match the dimensionality of LLMs. Model 4 only projects the [CLS] token into the LLM, which significantly reduces the computational cost of the LLM, achieving better results. These results highlight the effectiveness of the individual components designed within the PE-RSIC framework.

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TABLE VII
ABLATION RESULTS ON THE UCM DATASET. THE BOLD INDICATES THE BEST RESULTS

Method	Positions	Params(M)	Times(S)	BLEU-4	CIDEr	SPICE
Model8	Top	6.6	2.07	68.97	320.84	51.56
Model9	Bottom	6.6	2.07	72.43	378.67	53.47
Model10	0-8	8.1	2.10	73.01	385.34	54.61
Model4	All	9.6	2.16	**74.33**	**394.57**	**55.79**

4) Components of CRS-Adapter: As shown in Table VI, we conduct an ablation study by incrementally adding components of the CRS-Adapter. Model 5 represents adapter residual learning branch after removing the MSA module. Model 6 contains two full branches without gating mechanisms and weight-sharing. However, adding a residual learning branch still improves performance. Model 7 introduces weight-sharing only in the downsampling layer. Finally, Model 4 represents the complete CRS-Adapter architecture. Our experiments demonstrate the effectiveness of each component in CRS-Adapter.

5) Positions of CRS-Adapter and Speed Performance: As shown in Table VII, we evaluate the impact of inserting CRS-Adapters at different QFormer layers on model performance and inference efficiency. Adapters are inserted into: top six layers (Model8), bottom six layers (Model9), first nine layers (Model10), and all 12 layers (Model4). Performance improves with more inserted layers. Model4 achieves the best result, indicating that deeper integration helps better model the semantic alignment between RS images and text. In addition, Model9 clearly outperforms Model8 despite using the same number of adapters, suggesting that inserting adapters into deeper layers is more effective than in earlier ones. Regarding inference time, deeper insertion introduces only a minor overhead (from 2.07 to 2.16 s).

V. CONCLUSION

In this article, we propose PE-RSIC, a PETL framework tailored for RSIC. Our framework leverages the pretrained BLIP-2 model as the baseline and introduces two key components: the CRS-Adapter and the Class Prompt. The CRS-Adapter is specifically designed to adapt pretrained natural-domain vision–language models to the RS domain by learning cross-modal relationships in a lightweight manner. It is integrated in parallel within the QFormer blocks, using a dual-branch structure to retain query integrity and a gating mechanism to regulate the flow of information. Meanwhile, the Class Prompt, encoded from the [CLS] token of the vision encoder output, effectively captures geospatial class-specific features and guides the decoder in generating more accurate and context-aware captions for RS images. By freezing the parameters of the pretrained components and fine-tuning only the adapter and prompt modules, PE-RSIC achieves efficient domain adaptation with minimal computational and memory overhead, outperforming full fine-tuning methods with only 5% of the trainable parameters. Moreover, numerous advanced VLPs have emerged to better handle diverse tasks. Future work will focus on effectively integrating CRS-Adapter with other QFormer-based VLPs and further optimizing its memory footprint during both training and inference. We hope our work will inspire research toward a unified PETL framework for multimodal foundation models, addressing the diverse task and scenario requirements in RS.

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